

UniverSeg: Universal Medical Image Segmentation

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Abstract

While deep learning models have become the predominant method for medical image segmentation, they are typically not capable of generalizing to unseen segmentation tasks involving new anatomies, image modalities, or labels. Given a new segmentation task, researchers generally have to train or fine-tune models. This is time-consuming and poses a substantial barrier for clinical researchers, who often lack the resources and expertise to train neural networks.

We present **UniverSeg**, a method for solving unseen medical segmentation tasks without additional training. Given a query image and an example set of image-label pairs that define a new segmentation task, UniverSeg employs a new **CrossBlock mechanism to produce accurate segmentation maps without additional training**. To achieve generalization to new tasks, we have gathered and standardized a collection of **53 open-access medical segmentation datasets with over 22,000 scans**, which we refer to as **MegaMedical**. We used this collection to train UniverSeg on a diverse set of anatomies and imaging modalities. We demonstrate that Uni-

verSeg substantially outperforms several related methods on unseen tasks, and thoroughly analyze and draw insights about important aspects of the proposed system. The UniverSeg source code and model weights are freely available at <https://universeg.csail.mit.edu>

1. Introduction

Image segmentation is a widely studied problem in computer vision and a central challenge in medical image analysis. Medical segmentation tasks can involve diverse imaging modalities, such as magnetic resonance imaging (MRI), X-ray, computerized tomography (CT), and microscopy; different biomedical domains, such as the abdomen, chest, brain, retina, or individual cells; and different labels within a region, such as heart valves or chambers (Figure 1). This diversity has inspired a wide array of segmentation tools, each usually tackling one task or a small set of closely related tasks [17, 23, 41, 42, 87, 94]. In recent years, deep-learning models have become the predominant strategy for medical image segmentation [45, 74, 87].

A key problem in image segmentation is *domain shift*,

*Denotes equal contribution

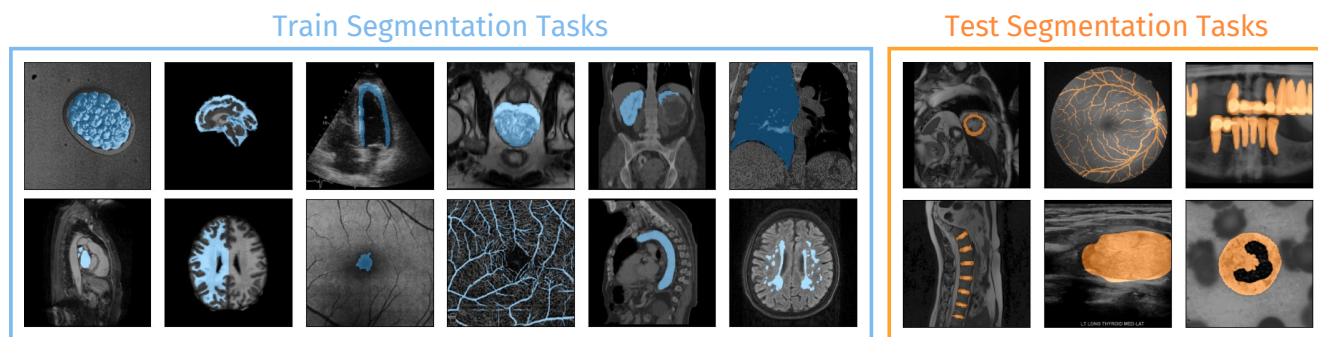


Figure 1: Medical segmentation involves many imaging types, biomedical domains, and target labels. We employ a large diverse set of training tasks (**blue**) to build a model that can segment unseen tasks (**orange**) without additional training.

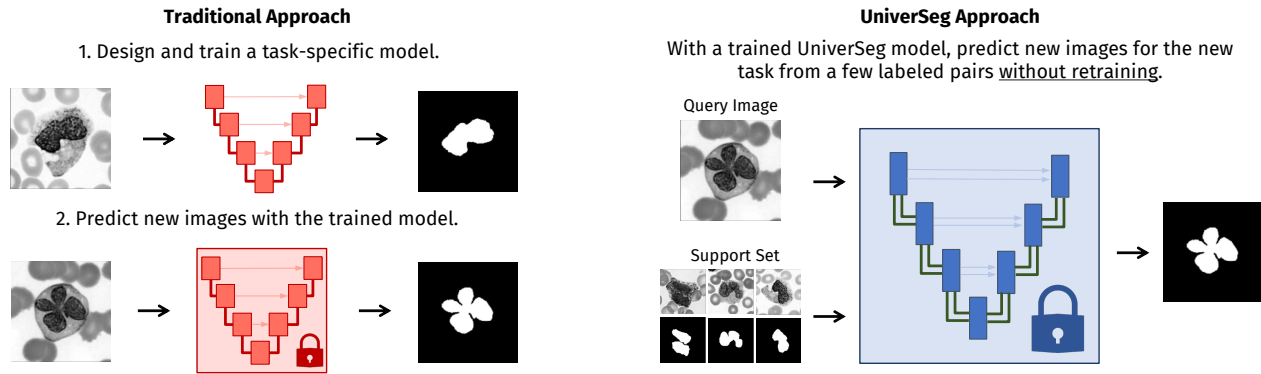


Figure 2: **Workflow for inference on a new task, from an unseen dataset.** Given a new task, traditional models (**left**) are trained before making predictions. UniverSeg (**right**) employs a *single* trained model which can make predictions for images (queries) from the new task with a few labeled examples as input (support set), without additional fine-tuning.

where models often perform poorly given out-of-distribution examples. This is especially problematic in the medical domain where clinical researchers or other scientists are constantly defining new segmentation tasks driven by evolving populations, and scientific and clinical goals. To solve these problems they need to either train models from scratch or fine-tune existing models. Unfortunately, training neural networks requires machine learning expertise, computational resources, and human labor. This is infeasible for most clinical researchers or other scientists, who do not possess the expertise or resources to train models. In practice, this substantially slows scientific development. We, therefore, focus on avoiding the need to do any training given a new segmentation task.

Fine-tuning models trained on the natural image domain can be unhelpful in the medical domain [86], likely due to the differences in data sizes, features, and task specifications between domains, and importantly still requires substantial retraining. Some few-shot semantic segmentation approaches attempt to predict novel classes without fine-tuning in limited data regimes, but mostly focus on classification tasks, or segmentation of new classes within the same input domain, and do not generalize across anatomies or imaging modalities.

In this paper, we present UniverSeg – an approach to learning a *single* general medical-image segmentation model that performs well on a variety of tasks without any retraining, including tasks that are substantially different from those seen at training time. UniverSeg learns how to exploit an input set of labeled examples that specify the segmentation task, to segment a new biomedical image in one forward pass. We make the following contributions.

- We propose UniverSeg – a framework that enables solving new segmentation tasks without retraining, using a novel flexible CrossBlock mechanism that transfers

information from the example set to the new image.

- We demonstrate that UniverSeg substantially outperforms several models across diverse held-out segmentation tasks involving unseen anatomies and even approaches the performance of fully-supervised networks trained specifically for those tasks.
- In extensive analysis, we show that the generalization capabilities of UniverSeg are linked to task diversity during training and image diversity during inference.

2. Related Works

Medical Image Segmentation. Medical image segmentation has been widely studied, with state-of-the-art methods training convolutional neural networks in a supervised fashion, predicting a label map for a given input image [23, 41, 42, 46, 87]. For a new segmentation problem, models are typically trained from scratch, requiring substantial design and tuning.

Recent strategies, such as the nnUNet [42], automate some design decisions such as data processing or model architecture but still incur substantial overhead from training. In contrast to these methods, UniverSeg generalizes to new medical segmentation tasks without training or fine-tuning.

Multi-task Learning. Multi-Task Learning (MTL) frameworks learn several tasks simultaneously [16, 24, 90]. For medical imaging, this can involve multiple modalities [75], population centers [64], or anatomies [76]. However, the tasks are always pre-determined by design: once trained, each network can only solve tasks presented during training. UniverSeg overcomes this limitation, enabling tasks to be dynamically specified during inference.

Transfer Learning. Transfer learning strategies involve fine-tuning pre-trained models, often from a different domain [66, 101]. This is used in medical image segmentation starting

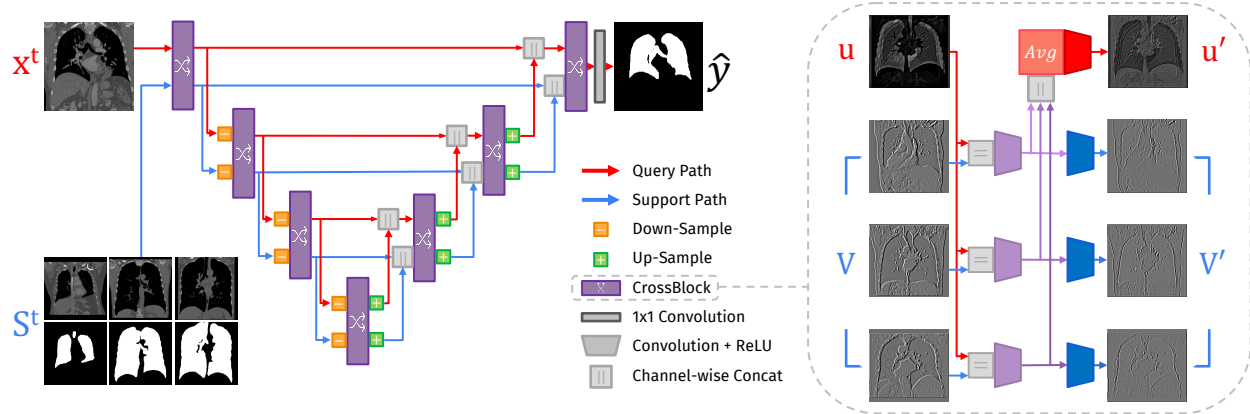


Figure 3: A UniverSeg network (**left**) takes as input a query image and a support set of image and label-maps (pairwise concatenated in the channel dimension) and employs multi-scale CrossBlock features. A CrossBlock (**right**) takes as input representations of the query u and support set $V = \{v_i\}$, and interacts u with each support entry v_i to produce u' and V' .

with models trained on natural images [4, 27, 44, 113, 116], where the amount of data far exceeds the amount in the target biomedical domain. However, **this technique still involves substantial training for each new task, which UniverSeg avoids**. Additionally, the differences between medical and natural images often make transfer learning from large pre-trained models unhelpful [86].

Optimization-based Meta-Learning. **Optimization-based meta-learning techniques** often learn representations that minimize downstream fine-tuning steps by using a few examples per task, sometimes referred to as few-shot learning [25, 78, 98, 104]. Meta-learning via fine-tuning has been studied in medical image segmentation to handle multiple image modalities [112], anatomies [110], and generalization to different targets [51, 52, 97]. While these strategies reduce the amount of data and training required for downstream tasks [33], fine-tuning these models nevertheless requires machine learning expertise and computational resources, which are often not available to medical researchers.

In-Context Learning. **In-Context Learning (ICL) methods adapt to new tasks without additional training by incorporating the task description as an input to the model** [14]. This strategy has been successfully demonstrated in both large language models [79, 108] and multi-modal foundation models which take interleaved text and images as inputs, maintaining natural language as the primary prompting mechanism [3]. Differing from recent image-based ICL image models which employ transformers [8, 106, 107], we develop a purely convolutional in-context learning method for medical image segmentation tasks, in which tasks are encoded as sets of image-label pairs.

Few-shot Semantic Segmentation. Few-shot (FS) methods adapt to new tasks from few training examples, often by

fine-tuning pretrained networks [25, 78, 104, 98]. Some few-shot semantic segmentation models generate predictions for new images (queries) containing unseen classes from just a few labeled examples (support) without additional retraining. One strategy prevalent in both natural image [77, 91, 109] and medical image [22, 62, 81, 95] FS segmentation methods is to employ large pre-trained models to extract deep features from the query and support images. These methods often involve learning meaningful prototypical representations for each label [105]. Self-supervised learning can help make up for the lack of training data and tasks [32, 80]. In contrast to UniverSeg, these methods focus on limited data regimes, tackle specific tasks, and generalize to new classes in a particular subdomain, like abdominal CT or MRI scans [32, 80, 88, 102]. **In our work, we focus on avoiding any fine-tuning, even when given many examples for a new task, to avoid requiring the clinical or scientific user to have machine learning expertise and computing resources.**

3. UniverSeg Method

Let t be a segmentation task comprised of a set of image-label pairs $\{(x_i^t, y_i^t)\}_{i=1}^N$. Common segmentation strategies learn parametric functions $\hat{y} = f_\theta^t(x)$, where f_θ^t is most often modeled using a convolutional neural network that estimates a label map $\hat{y}$ given an input image x . By construction, f_θ^t only learns to predict segmentations for task t .

In contrast, we learn a universal function $\hat{y} = f_\theta(x^t, S^t)$ that predicts a label map for input x^t of task t , according to the task-specifying support $S^t = \{(x_j^t, y_j^t)\}_{j=1}^n$ comprised of example image-label pairs available for t .

3.1. Model

We implement f_θ using a fully convolutional neural network illustrated in Figure 3. We first introduce the pro-

posed building blocks: the *cross-convolution* layer and the *CrossBlock* module. We then specify how we combine these blocks into a complete segmentation network.

CrossBlock. To transfer information between the support set and query image, we introduce a *cross-convolution* layer that interacts a query feature map u with a set of support feature maps $V = \{v_i\}_{i=1}^n$:

$$\text{CrossConv}(u, V; \theta_z) = \{z_i\}_{i=1}^n, \quad (1)$$

for $z_i = \text{Conv}(u \parallel v_i; \theta_z)$,

where $\parallel$ is the concatenation operation along the feature dimension and $\text{Conv}(x; \theta_z)$ is a convolutional layer with learnable parameters θ_z . Due to the weight reuse of θ_z , cross-convolution operations are permutation invariant with respect to V . From this layer, we design a higher-level building block that produces updated versions of query representation u and support V at each step in the network:

$$\begin{aligned} \text{CrossBlock}(u, V; \theta_z, \theta_u, \theta_v) &= (u', V'), \text{ where:} \quad (2) \\ z_i &= \phi(\text{CrossConv}(u, v_i; \theta_z)) \quad \text{for } i = 1, 2, \dots, n \\ u' &= \phi(\text{Conv}(1/n \sum_{i=1}^n z_i; \theta_u)) \\ v'_i &= \phi(\text{Conv}(z_i; \theta_v)) \quad \text{for } i = 1, 2, \dots, n, \end{aligned}$$

where $\phi(x)$ is a non-linear activation function. This strategy enables the representations of each support set entry and query to interact with the others through their average representation, and facilitates variably sized support sets.

Network. To integrate information across spatial scales, we compose the CrossBlock modules in an encoder-decoder structure with residual connections, similarly to the popular UNet architecture (Figure 3). The network takes as input the query image x^t and support set $S^t = \{(x_i^t, y_i^t)\}_{i=1}^n$ of image and label-map pairs, each concatenated channel-wise, and outputs the segmentation prediction map $\hat{y}^t$.

Each level in the encoder path consists of a CrossBlock followed by a spatial down-sampling operation of both query and support set representations. Each level in the expansive path consists of up-sampling both representations, which double their spatial resolutions, concatenating them with the equivalently-sized representation in the encoding path, followed by a CrossBlock. We perform a single 1x1 convolution to map the final query representation to a prediction.

3.2. Training

Algorithm 1 describes UniverSeg training using a large and varied set of training tasks $\mathcal{T}$ and the loss

$$\mathcal{L}(\theta; \mathcal{T}) = \mathbb{E}_{t \in \mathcal{T}} \mathbb{E}_{(x^t, y^t), S^t} [\mathcal{L}_{\text{seg}}(f_\theta(x^t, S^t), y^t)], \quad (3)$$

where $x^t \notin S^t$, and $\mathcal{L}_{\text{seg}}(\hat{y}, y^t)$ is a standard segmentation loss like cross-entropy or soft Dice [74], capturing the agreement between the predicted $\hat{y}$ and ground truth y_t .

Data Augmentation. We employ data augmentation to grow

Algorithm 1 UniverSeg Training Loop using SGD with learning rate η over tasks $\mathcal{T}$, main architecture f_θ , in-task augmentations Aug_t and task augmentations Aug_T

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for  $k = 1, \dots, \text{NumTrainSteps}$  do
   $t \sim \mathcal{T}$  ▷ Sample Task
   $(x_i^t, y_i^t) \sim t$  ▷ Sample Query
   $S^t \leftarrow \{(x_j^t, y_j^t)\}_{j \neq i}^n$  ▷ Sample Support
   $x_i^t, y_i^t \leftarrow \text{Aug}_t(x_i^t, y_i^t)$  ▷ Augment Query
   $S^t \leftarrow \{\text{Aug}_t(x_j^t, y_j^t)\}_j^n$  ▷ Augment Support
   $x_i^t, y_i^t, S^t \leftarrow \text{Aug}_T(x_i^t, y_i^t, S^t)$  ▷ Task Aug
   $\hat{y}_i \leftarrow f_\theta(x_i^t, S^t)$  ▷ Predict label map
   $\ell \leftarrow \mathcal{L}_{\text{seg}}(\hat{y}_i, y_i^t)$  ▷ Compute loss
   $\theta \leftarrow \theta - \eta \nabla_\theta \ell$  ▷ Gradient step
end for

```

the diversity of training tasks and increase the number of effective training examples belonging to any particular task.

In-Task Augmentation – $\text{Aug}_t(x, y)$. To reduce overfitting to individual subjects, we perform standard data augmentation operations, like affine transformations, elastic deformation, or adding image noise to the query image and *each* entry of the support set independently.

Task Augmentation – $\text{Aug}_T(x, y, S)$. Similar to standard data augmentation that reduces overfitting to training examples, augmenting the training *tasks* is useful for generalizing to *new tasks*, especially those far from the training task distribution. We introduce task augmentation – alterations that modify all query and support images, and/or all segmentation maps, with the same type of task-changing transformation. Example task augmentations include edge detection of the segmentation maps or a horizontal flip to all images and labels. We provide a list of all augmentations and the parameters we used in the supplemental Section C.

3.3. Inference

For a given query image x^t , UniverSeg predicts segmentation $\hat{y} = f_\theta(x^t, S^t)$ given a support set S^t , where the prediction quality depends on the choice of the support set S^t . To reduce this dependence, and to take advantage of more data when memory constraints limit the support set size at inference, we combine predictions from an ensemble of K independently sampled support sets $\{S_i^t\}_{i=1}^K$ as their pixel-wise average to produce the prediction $\hat{y} = \frac{1}{K} \sum_{k=1}^K f_\theta(x, S_k^t)$.

4. MegaMedical Dataset

To train our universal model f_θ , we employ a set of *segmentation tasks that is large and diverse, so that it is able to generalize to new tasks*. We compiled MegaMedical – an extensive collection of open-access medical segmentation datasets with diverse anatomies, imaging modalities, and

labels. It is constructed from 53 datasets encompassing 26 medical domains and 16 imaging modalities.

We standardize data across the wildly diverse formats of original datasets, processed images, and label maps. We also expand the training data using synthetic segmentation tasks to further increase the training task diversity. Because of individual dataset agreements, we are prohibited from re-releasing our processed version of the datasets. Instead, we will provide data processing code to construct MegaMedical from its source datasets.

Datasets. MegaMedical features a wide array of biomedical domains, such as eyes [40, 61, 69, 84, 99], lungs [89, 93, 96], spine vertebrae [114], white blood cells [115], abdominal [11, 13, 35, 43, 49, 57, 58, 60, 63, 67, 68, 85, 96], and brain [5, 28, 36, 55, 56, 70, 71, 72, 96], among others. Supplemental Table 3 provides a detailed list of MegaMedical datasets. Acquisition details, subject age ranges, and health conditions are different for each dataset. We provide data processing details in supplemental Section A.

Medical Image Task Creation. While datasets in MegaMedical feature a variety of imaging tasks and label protocols, in this work we focus on the general problem of 2D binary segmentation. For datasets featuring 3D data, for each subject, we extract the 2D mid-slice of the volume along all the major axes. When multiple modalities are present, we include each modality as a new task. For datasets containing multiple segmentation labels, we create as many binary segmentation tasks as available labels. All images are resized to 128×128 pixels and intensities are normalized to the range $[0,1]$.

Synthetic Task Generation. We adapt the image generation procedure involving random synthetic shapes described in SynthMorph [37] to produce a thousand synthetic tasks to be used alongside the medical tasks during training. We detail the generation process and include examples of synthetic tasks in supplemental Section D.

5. Experiments

We start by describing experimental details. The first set of experiments compares the performance of UniverSeg in the held-out datasets against several single-pass methods used in few-shot learning. We then report on a variety of analyses, including ablations of modeling decisions, and the effect of training task diversity, support set size, and number of examples available for a new task.

5.1. Experimental Setup

Model. We implement the network in UniverSeg (Figure 3) using an encoder with 5 CrossBlock stages and a decoder with 4 stages, with 64 output features per stage and LeakyReLU non-linearities after each convolution. We use

bilinear interpolation when downsampling or upsampling.

Data. For each dataset d , we construct three disjoint splits $d = \{d_{\text{support}}, d_{\text{dev}}, d_{\text{test}}\}$ with 60%, 20%, and 20% of the subjects, respectively. Similar to dataset generalization [103], we divide the available datasets into a training set $\mathcal{D}^T$ and a held-out test set $\mathcal{D}^H$. We train models using the support and development splits of the training datasets $\{d_{\text{support}} | d \in \mathcal{D}^T\}$. We performed model selection and hyper-parameter tuning using the development split of held-out dataset WBC, and trained models until they stopped improving in the d_{dev} split, averaged across the held-out datasets. We report results using the unseen test split of the held-out datasets $\{d_{\text{test}} | d \in \mathcal{D}^H\}$. Support set image-label pairs are sampled with replacement from each dataset’s support split.

For held-out datasets, we evaluated three datasets containing anatomies represented in the training datasets (ACDC [10] and SCD [85] (heart), and STARE[40] (retinal blood vessels)), and three datasets of anatomies not covered by the rest of MegaMedical (PanDental [2] (mandible), SpineWeb [114] (vertebrae), and WBC [115] (white blood cells)).

Few-Shot Baselines. We compare UniverSeg models to three segmentation methods from the few-shot (FS) literature, since these approaches also predict the segmentation of a query image given a support set of image-label pairs, although they were designed for the low-data regime. SE-net [88] features a fully-convolutional network, squeeze-excitation blocks, and a UNet-like model architecture. ALP-Net [80] and PANet [105], employ prototypical networks that extract prototypes from their inputs to match the given query with the support set. While ALPNet also employs a self-supervised method to generate additional label maps in settings with few tasks, we omit this step since MegaMedical includes a large collection of tasks.

Unlike UniverSeg, these methods were designed to generalize to similar tasks, such as different labels in the same anatomy and image type, or different modalities for the same anatomy. To make the comparison to UniverSeg fair, we make several additions to the training and inference procedures of these baselines as described below, and chose the best performing variant of each baseline.

Supervised Task-Specific Models. While it is often impractical for clinical researchers to train individual networks for each task, for evaluation we train a set of task-specific networks to serve as an upper bound of supervised performance on the held-out datasets. We employ the widely-used nnUNet [42], which automatically configures the model and training pipeline based on data properties. Each model is task-specific, using the support and development splits for training and model selection, respectively. We report results on the test split.

Model	#Params	Runtime ms	Dice Score
PANet	14.71	240.0 $\pm$ 1.8	41.8 $\pm$ 1.3
ALPNet	43.02	527.7 $\pm$ 8.7	47.8 $\pm$ 1.1
SENet	0.92	4.1 $\pm$ 0.8	50.1 $\pm$ 1.3
UniverSeg (ours)	1.18	142.0 $\pm$ 0.4	71.8 $\pm$ 0.9
nnUNet (sup.)	17 $\times$ 1.87	17 $\times$ 1.4 $\cdot$ 10 ⁷	84.4 $\pm$ 1.0

Table 1: **Performance Summary.** For UniverSeg and each FS baseline we report model size (in millions), inference runtime, and average held-out Dice score (with bootstrapping standard deviation). As an upper bound, we include the set of 17 individually trained task-specific nnUNets for the 6 held-out datasets, where their run-time is their cumulative required training time.

Evaluation. We evaluate models on the held-out datasets $\mathcal{D}^H$ using the test split for query images and the support split for support-sets. For all methods, unless specified otherwise, we perform 5 independent predictions per test subject using randomly drawn support sets, and ensemble the predictions. We enforce that the same random support sets are used for all methods. We evaluate predictions using the Dice score [21] (0 - 100, 0=no overlap, 100=perfect match), which quantifies the region overlap between two regions and is widely used in medical segmentation. For tasks with more than one label, we average Dice across all labels. For datasets with multiple tasks, we average performance across all tasks. We estimate prediction variability using subject bootstrapping, with 1,000 independent repetitions. At each repetition, we treat each task independently, sampling subjects with replacement, and report the standard deviation across bootstrapped estimates.

Training. We train networks with the Adam optimizer [53] and soft Dice loss [74, 100]. For the ALPNet and PANet baselines, we add a prototypical loss term as described in their original works. Models trained with cross-entropy performed substantially worse than soft Dice.

While the original baseline methods were not introduced with significant data augmentation, we trained all UniverSeg and FS models with and without the proposed augmentation transformations, and report results on the best-performing setting. Unless specified otherwise, models are trained using a support size of 64. While the baselines were originally designed with small support sizes (1 or 5) as they tackled the few-shot setting, we found that training and evaluating them with larger support sizes improved their performance.

Implementation. We provide additional implementation and experimental details in supplemental Section B. Code and pre-trained model weights for UniverSeg are available at <https://universeg.csail.mit.edu>.

5.2. Task Generalization Results

First, we compare the segmentation quality of UniverSeg with FS baselines and the task-specific upper bounds. Our primary goal is to assess the effectiveness of UniverSeg in solving tasks from unseen datasets. Figure 4 presents the average Dice scores per dataset for each method, and Figure 5 presents example segmentation results for each method and dataset.

Few-shot methods. UniverSeg significantly outperforms all FS methods in all held-out datasets. For each FS method, we report the best-performing model, which involved adding components of the UniverSeg training pipeline. In the supplemental material, we show that few-shot methods perform worse when trained with a support set size of 1 and without ensembling, as they were originally introduced.

UniverSeg outperforms the highest performing baseline for all datasets with Dice improvements ranging from 7.3 to 34.9. Figure 5 also shows clear qualitative improvements in the predicted segmentations. Given the similarities between SENet and UniverSeg (fully convolutional UNet-like structure), these results suggest that the proposed CrossBlock is better suited to transferring spatial information from the support set to the query. Table 1 shows that UniverSeg also requires fewer model parameters than PANet, ALPNet, and the nnUNets, and a similar number to SENet.

Task-specific networks. For some datasets like PanDental or WBC, UniverSeg performs competitively with the supervised task-specific networks, which were extensively trained on each of the held-out tasks, and are unfeasible to run in many clinical research settings. Moreover, from the qualitative results of Figure 5, we observe that segmentations produced by UniverSeg more closely match those of the supervised baselines than those of any other few-shot segmentation task, especially in challenging datasets like SpineWeb or STARE.

5.3. Analysis

We analyze how several of the data, model, and training decisions affect the performance of UniverSeg.

Task Quantity and Diversity. We study the effect of the number of datasets and individual tasks used for training UniverSeg. We leave out synthetic tasks for this experiment, and train models on random subsets of the MegaMedical training datasets.

Figure 6 presents performance on the held-out datasets for different random subsets of training datasets. We find that having more training tasks improves the performance on held-out tasks. In some scenarios, the *choice* of datasets has a substantial effect. For instance, for models trained with 10% of the datasets, the best model outperforms the worst one by 17.3 Dice points, and comparing those subsets

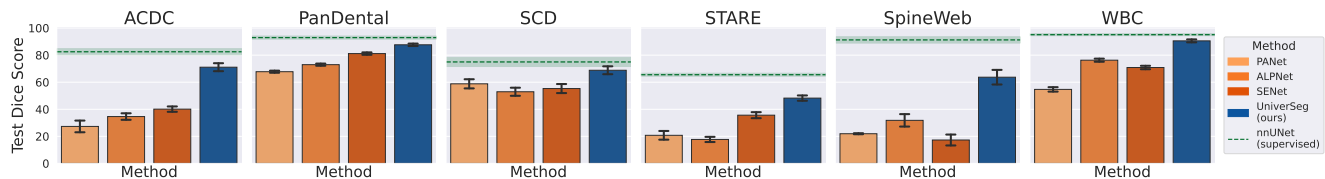


Figure 4: **Average Dice score per each held out dataset.** Performance of UniverSeg and several few-shot baselines, and the upper bound of each dataset determined by the individual fully-trained networks. For each of the unseen datasets, we average across tasks and subjects, and show the bootstrap variability in the error bars.

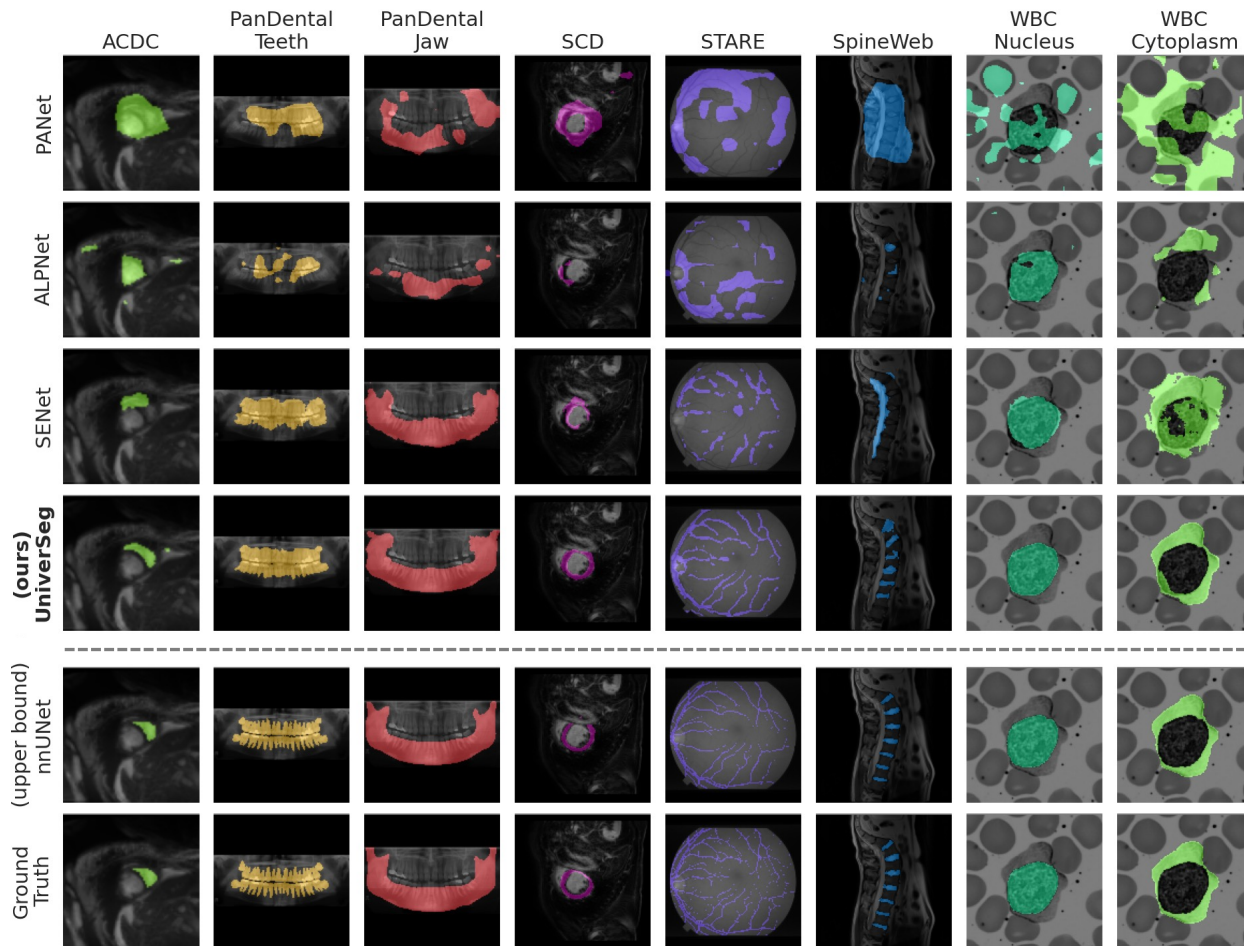


Figure 5: **Example model predictions for unseen tasks.** For a randomly sampled image per held-out task, we visualize the predictions of UniverSeg, few-shot baselines, and individually trained nnUNet models, along with ground truth maps.

we find that the best performing one was trained on a broad set of anatomies including heart, abdomen, brain, and eyes; while the least accurate model was trained on less common lesion tasks, leading to worse generalization.

Ablation of Training Strategies. We perform an ablation study over the three main techniques we employ for increasing data and task diversity during training: in-task augmentation, task augmentation, and synthetic tasks.

Table 2 shows that all proposed strategies lead to improve-

ments in model performance, with the best results achieved when using all strategies jointly, providing a boost of 9 Dice points over no augmentations or synthetic tasks. Incorporating task augmentation leads to the largest individual improvement of 7.7 Dice points. Remarkably, the model trained using only synthetic data performs surprisingly well on the medical held-out tasks despite having never been exposed to medical training data. These results suggest that increasing image and task diversity during training, even artificially, has

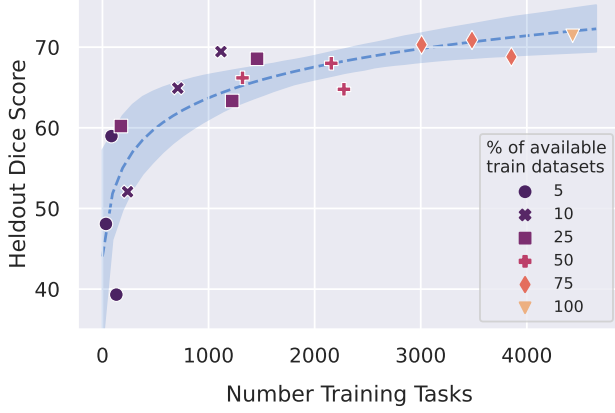


Figure 6: **Average held-out Dice versus the number of training tasks.** Points represent individual UniverSeg networks trained on a percentage of available training datasets and shown in terms of the number of underlying training tasks. In blue, we report a logarithmic fit to the data and 95% confidence intervals obtained by bootstrapped fits.

Synth	Medical	In-Task	Task	Dice Score
✓				61.7 ± 1.5
	✓			62.7 ± 1.1
✓	✓			64.5 ± 1.0
	✓	✓		67.0 ± 0.9
	✓		✓	70.4 ± 1.3
	✓	✓	✓	70.0 ± 1.5
✓	✓	✓	✓	71.8 ± 0.9

Table 2: **Training Strategies Ablation.** Average held-out Dice for UniverSeg models trained with different combinations of proposed techniques to increase task diversity: in-task augmentation, task augmentation, and synthetic tasks.

a substantial effect on how the model generalizes to unseen segmentation tasks.

Support Set Size. We study the effect of support size on models trained with support sizes N from 1 to 64.

Figure 7 shows that the best results are achieved with large training support set sizes, with the average held-out Dice rapidly improving from 53.7 to 69.9 for supports sizes from 1 to 16, and then providing diminishing returns at greater support sizes, with a maximum of 71 Dice at support size 64. We find that ensembling predictions leads to consistent improvements in all cases, with greater improvements of 2.4-3.1 Dice points for small support sets ($N < 16$).

Limited Example Data. Since manually annotating examples from new tasks is expensive for medical data, we investigate how the number of labeled images affects the performance of UniverSeg. We study UniverSeg when using a limited amount of labeled examples N at inference, for

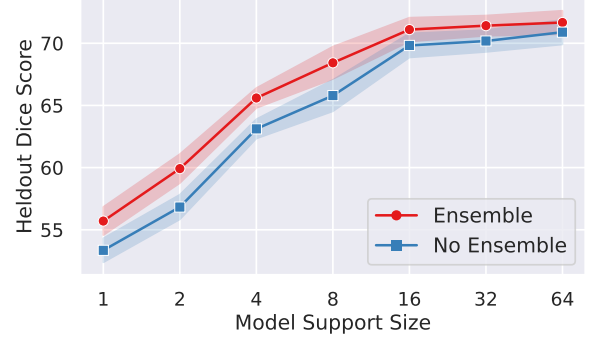


Figure 7: **Effect of support size.** Relationship between models trained at certain support sizes and their average held-out Dice score. Results improve with higher support size, with ensembling consistently helping.

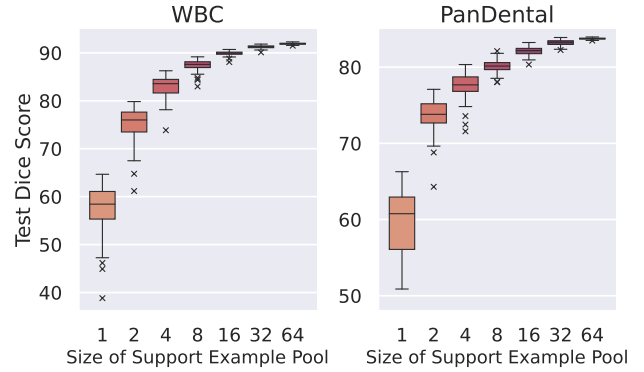


Figure 8: **Effect of available data at inference.** UniverSeg predictions using a limited d_{support} example pool on the held-out WBC and PanDental datasets. For each size, we perform 100 repetitions using different random subsets.

$N = 1, 2, \dots, 64$. We perform 100 repetitions for each size, each corresponding to an independent random subset of the data. Here, the support set contains all available data for inference, and thus we do not perform ensembling.

Figure 8 presents results for the WBC and PanDental held-out datasets, which have 108 and 116 examples in their d_{support} splits respectively. For small values of support size N , we observe a large variance caused by very diverse support sets. As N increases, we observe that average segmentation quality monotonically improves and the variance from the sample of available data examples is greatly reduced.

Support Set Ensembling. We study the effect of varying the support size N at inference, and number K of predictions being ensembled. We first sample 100 independent support sets for each inference support size N . Then, for each ensembling amount K , we compute ensembled predictions by averaging K independently drawn predictions.

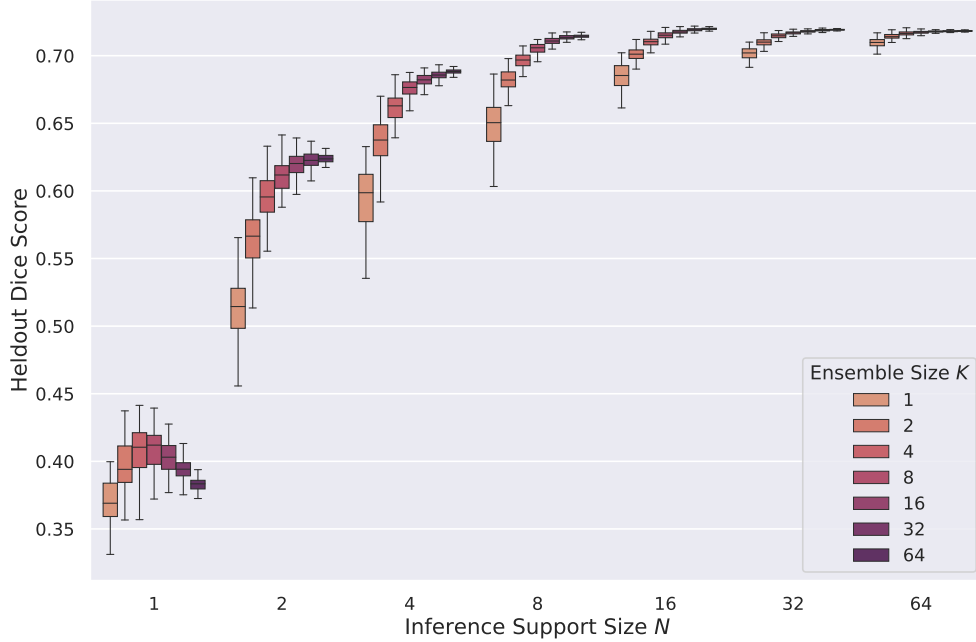


Figure 9: **Ensembling predictions at different inference support sizes.** Average held-out test Dice Score for different settings of ensembling and support size. For each inference support size N , we report the results (in average held-out Dice Score) of taking 100 predictions ($K = 1$) and ensembling by averaging in groups of size K , performing 100 repetitions for each K . The value boxes report quantiles over the 100 values for each setting and find that increasing either K or N leads to improved model performance, with N having a significantly larger effect than K .

Figure 9 shows that given a certain support size, increasing the ensemble size leads to monotonic improvements and reduced variance, likely by being less dependent on the specific examples in the support set. The performance also monotonically improves with increased support size N , which has a significantly larger effect on segmentation accuracy than increasing the ensemble size. For instance, non-ensembled predictions with support size 64 ($N = 64$, $K = 1$) are better than heavily ensembled predictions with smaller support sizes ($N = 2, 4, 8$ and $K = 64$), even though the latter uses more support examples. This suggests that UniverSeg models exploit information coming from the support examples in a fundamentally different way than existing ensembling techniques used in FS learning.

6. Discussion and Conclusion

We introduce UniverSeg, an approach for learning a *single* task-agnostic model for medical image segmentation. We use a large and diverse collection of open-access medical segmentation datasets to train UniverSeg, which is capable of generalizing to unseen anatomies and tasks. UniverSeg introduces the idea that segmentation tasks from diverse biomedical domains can be defined, or prompted, by a set of segmentation examples. We introduce a novel *cross-convolution* operation that interacts the query and support representations

at different scales.

In our experiments, UniverSeg substantially outperforms existing few-shot methods in all held-out datasets. Through extensive ablation studies, we conclude that UniverSeg performance is strongly dependent on task diversity during training and support set diversity during inference. This highlights the utility of UniverSeg facilitating variably-sized support sets, enabling flexibility to potential users’ datasets.

Limitations. In this work, we focused on demonstrating and thoroughly analyzing the core idea of UniverSeg, using 2D data and single labels. We are excited by future extensions to segment 3D volumes using 2.5D or 3D models and multi-label maps, and further closing the gap with the upper bounds.

Outlook. UniverSeg promises to easily adapt to new segmentation tasks determined by scientists and clinical researchers, without model retraining which is often impractical for them.

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